Month 8 Review and Recovery Plan

ARES
CONSORTIUM

Solis

THARSIS

Meridian

helix

Project Status

Four things have gone wrong at month 8, and none is a failure of effort. All four follow from one document nobody has written: an interface contract saying what crosses between the four partners' models, in what units, on which clock, owned by whom.

Solis and Tharsis have yet to exchange anything on time.

Solis sizes generating capacity over multi-year horizons, Tharsis simulates habitat demand hour by hour, and their one exchange is two months late and in a format the receiver did not plan for. Nobody wrote down when it was due or what it should contain.

Tharsis and Helix disagree about what the shared variables mean and who owns them.

Helix is building the translation layer between the partners' models. With no owner named for those variables, it read them its own way and built its tooling on that reading.

Meridian says the interface cannot carry a safety claim.

Meridian is the risk laboratory that sets what counts as acceptable service and judges the evidence. It needs every exchanged number to arrive with a stated uncertainty, and none of them do.

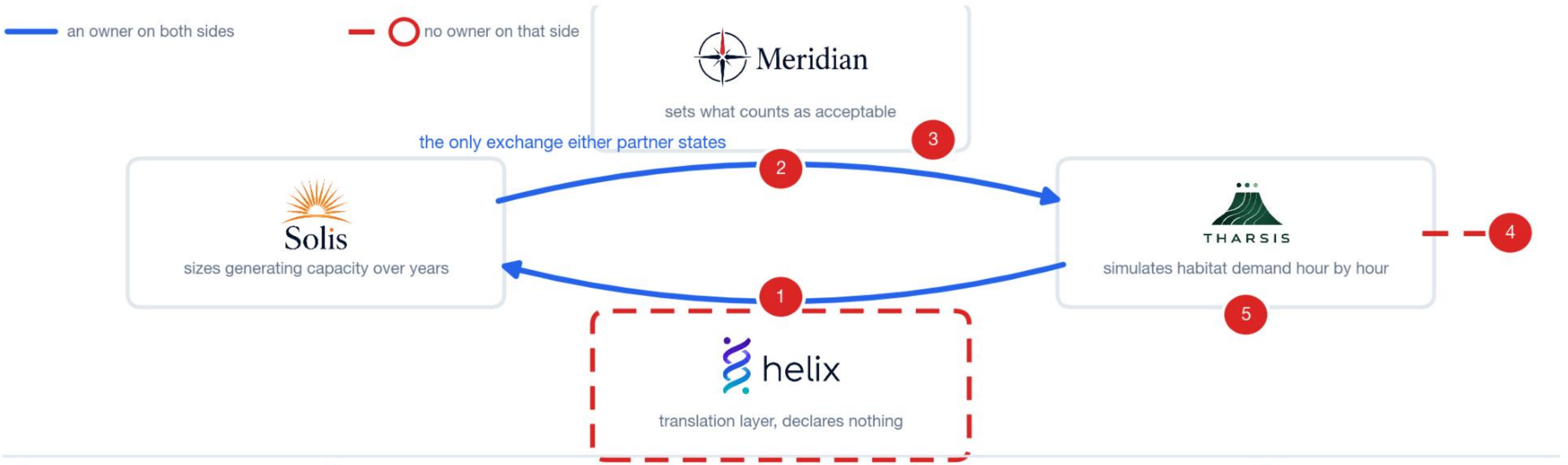
Nobody has shown the two models have to run together.

The conversions between Tharsis's hourly demand and Solis's multi-year capacity plan belong to no partner, so the coupling has never been tested.

Second-year funding depends on one end-to-end demonstration, and there is no demonstration.

Ownership Gap

Solis sizes generation over years, Tharsis simulates demand hour by hour, Meridian sets what counts as acceptable. Five things their models need that no partner produces. Of six possible partner pairs, only one states any exchange at all.



1. **Turning hourly demand into a planning bound**
   Tharsis emits a value every hour. Solis plans against one number. Nobody converts.
2. **Turning installed capacity into available capacity**
   Solis publishes what it builds. Tharsis needs what survives a storm, so invents a rule.
3. **Thresholds and stress scenarios**
   Meridian produces both and sends them to nobody.
4. **The population schedule**
   Drives every demand profile. No partner produces it, nobody versions it.
5. **Which loads can be shed**
   A storm collapses supply. The load keeping people alive does not move. Nobody owns the split.

None of these is a disagreement.
Each is work nobody agreed to do.

Somebody is doing them privately
and writing nothing down.

Model Fidelity

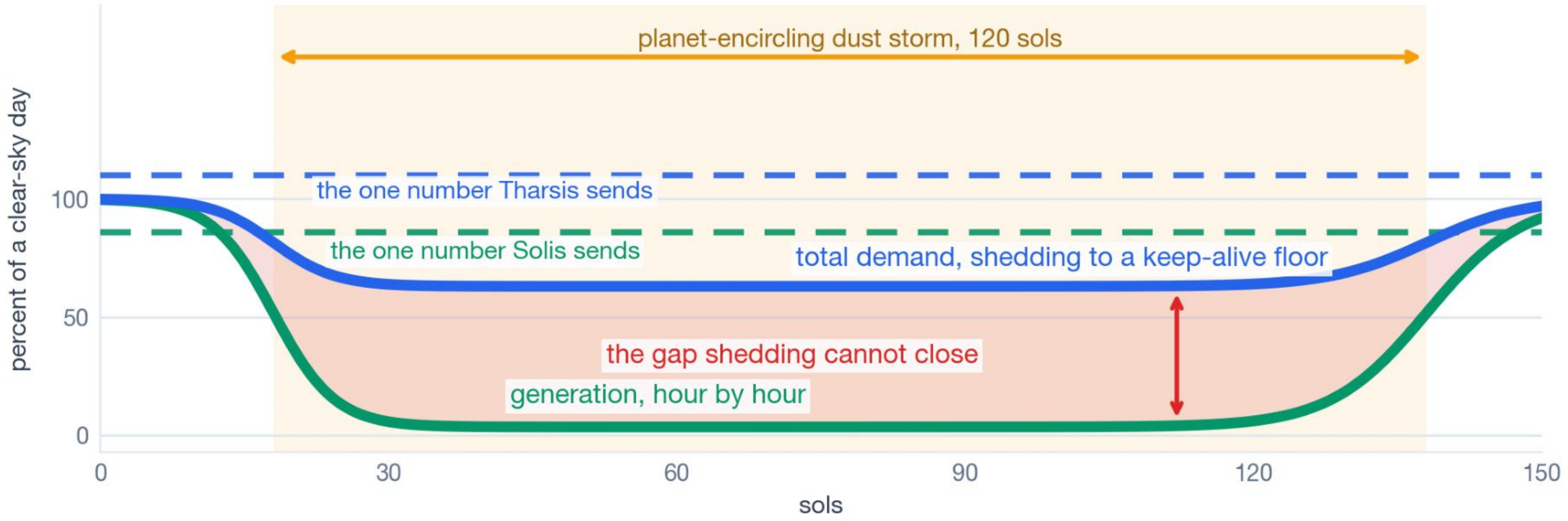
Tight coupling between the two models is a requirement, not a preference. One storm moves both sides at once, generation and demand answering to the same weather.

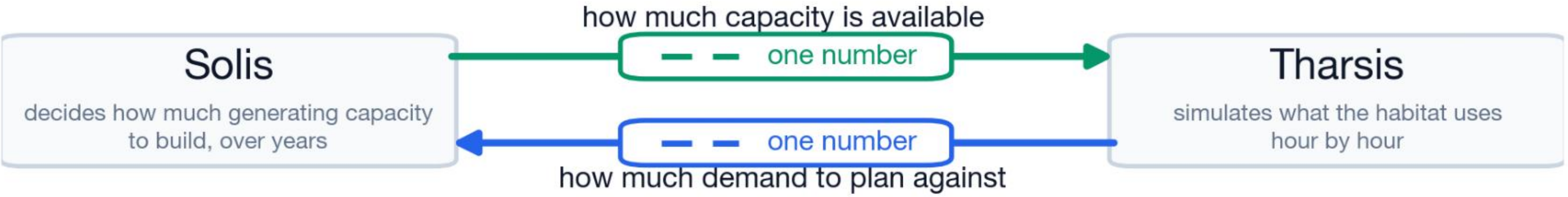
The shed is published, not assumed.
NASA surface power tables hold the habitat flat and shed rover and drill to zero, keep-alive 22 kW against 35 kW normal.

The deciding gap is which loads can be shed.
No partner owns that split, so neither can compute the floor supply must still meet.

The interface sends one bound, so the two effects arrive separately.
Each model sees a worst case for its own side, never both at once.

Somebody signs a habitat design on the strength of this model.
Below some level of detail it is not weak evidence for that decision, it is no evidence, and Meridian, the risk laboratory, has already said the interface cannot support the risk claims made from it.





Delivery Approach

Week 2 produces something that runs. Every week after that improves it.

Week 2, crude on purpose

One sol, nothing polished, so every format and calendar mismatch fails here while there is still time. Month 8, and the consortium has never run the two models end to end.

Week 5, slack that also tests the case

Solis revises its design after seeing the stress run. If it barely changes, the feedback between the models is weak and tight coupling was not needed.

The usual plan fails late.
Integrating at the end finds the same mismatches in week 6, with no time left.

Second year funding depends on this.
The board sees a demonstration, not a plan for one.

Six Week Plan

Six weeks, planned so that a week that goes wrong changes the plan rather than the date.

Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
3 sessions	3 sessions	2 sessions	1 session	1 session	1 session
Contract v0	End to end on junk data	Contract v1	Normal case runs	Traceability pass	Board demonstration
Calendar fixed, thresholds published, schedule owned.	Every format and encoding mismatch fails here.	Complete, or with its empty cells visible.	One configuration, traceable.	If week 4 held. Catch-up if it did not.	Normal case and one stress event.

Solis to Tharsis, the exchange that already failed
Nothing else runs until this exchange works.

CRITICAL PATH

BUFFER
Nothing scheduled here that cannot move out.

Meridian, the risk laboratory
Runs alongside, because what Meridian requires changes what Solis to Tharsis must carry.

Run it
Normal case, stress case, traceability.

Slack and rehearsal
One week absorbed, then four days of polish.

Helix conformance check
The re-pointed adapter runs on every exchange from week 2.

WHAT HAS TO BE TRUE WHEN THE WEEK ENDS

Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Calendar fixed and thresholds published, or week 2 carries an unowned assumption.	It ran end to end, however badly. If it did not, that is the finding and the plan changes.	Contract table complete or visibly incomplete. An empty cell is acceptable, an unnoticed one is not.	The demonstration is better than last week's and still runs. That is the weekly test from here.	Stress case has run. If not, week 5 absorbs it and week 6 still rehearses.	The same run, presented, four days before the board sees it.

One slip is affordable, two are not.
Week 5 is all the slack there is.

This holds only if each partner can be required to write its side down.
That is the third decision, at the end of this deck.

Meeting Plan

Every decision is made in the first three weeks. Weeks 4 to 6 review work already agreed.

WEEK 1
First written agreement

Kickoff 90 min
State the diagnosis. The written agreement is the object, not the vocabulary. No ruling here.
Solis Tharsis Meridian Helix

Interface working session half day
Calendar, epoch, units, encoding, each owned. First agreement leaves the room.
Solis Tharsis Meridian Helix

Evidence elicitation 90 min
Meridian states what makes a resilience claim credible, and publishes the thresholds Tharsis lacks.
Solis Tharsis Meridian Helix

WEEK 2
End to end on junk data

Derate session 90 min
Solis states how capacity degrades. Meridian defines failure. The derate leaves Tharsis.
Solis Tharsis Meridian Helix

Agreement sign-off half day
Every cell filled or visibly empty. Helix re-points its adapter to the ruling, a diff not a rewrite.
Solis Tharsis Meridian Helix

Walking skeleton review 60 min
It ran end to end on junk data. Every mismatch fails here, in time to fix.
Solis Tharsis Meridian Helix

WEEK 3
Agreement complete

Stress case selection 60 min
Meridian names the one material stress event, its duration and what it stresses. Nothing else in the plan produces one, and week 5 runs whatever is chosen here, so this choice fixes what the demonstration can claim under load.
Solis Tharsis Meridian Helix

Integration standup 30 min weekly
Blockers only, and nothing is settled here. Solis, Tharsis and Helix keep the exchange running between the reviews, once a week through weeks 3, 4 and 5.
Solis Tharsis Meridian Helix

WEEK 4
Normal case runs

Normal case review 90 min
The normal operating case runs end to end with the agreement enforced, on the agreed season, dust cycle point and population state. From here the weekly test is that the demonstration is better than last week and still runs.
Solis Tharsis Meridian Helix

WEEK 5
Traceability pass

Stress case review 90 min
Meridian's named event, run iterated. Solis proposes a configuration, Tharsis simulates it, Solis revises. The size of that revision, against the tolerance set in week 1, is the test of whether tight coupling is needed at all.
Solis Tharsis Meridian Helix

WEEK 6
Board demonstration

Dress rehearsal half day
The traceability pass and the review pack, rehearsed four days before the board sees it. For every number in the result: which model produced it, against which declared agreement, on which version of the population schedule, under which threshold set.
no attendee list in the plan

Partner in the room · Partner holds the ruling · Partner not in the room · this session settles something

Week 5 absorbs slippage.
Nothing scheduled into it cannot move out of it.

Only the derate session cannot slip.
If it does, week 4 runs on an assumption nobody owns.

Partner Alignment

No partner needs a new incentive to sign an interface contract. Each is already paying for the absence of one. Two of those costs are on the month 8 record, two are argued from it, and the slide says which is whichters.

Helix Dynamics, building the schema translation layer between the models, is the clearest case.
It has already built tooling against a provisional reading of interface variables whose meaning and ownership it and Tharsis still dispute. If that reading is wrong, the tooling is rework.

Meridian Risk Laboratory, which defines failure conditions and acceptable service levels, has already asked for this.
It states that the interface cannot represent failure or uncertainty well enough to support credible mission-risk claims. Its own product, the risk judgement, is what the gap blocks.

Solis Systems, sizing generation and storage capacity over multi-year horizons, is exposed on the record.
Its one exchange to Tharsis is two months late and the earliest output is not in the format the receiver planned around. Nothing on file says what that output owed or when, so it can miss the same way twice.

Tharsis Habitat Institute, the university team simulating habitat demand hour by hour, is the receiver.
Its inputs are capacity and operating constraints another partner produces, so the delay is time it cannot recover by working harder, and it planned around a format that never arrived.

The first two are stated in the record. The last two are argued, from the late exchange and from the scale mismatch nobody has yet shown a coupling can bridge. That mismatch is what Solis and Tharsis share, and the contract settles what has to cross at all.

Decisions Required

Four decisions. Three change only what the integration coordinator may ask for. The fourth narrows what week six delivers.

Endorse the diagnosis.

Five things the models need that no partner produces, two of them the exchange between Solis's capacity plan and Tharsis's hourly simulation.

Back the sequence.

Solis to Tharsis first, with the risk laboratory Meridian setting its evidence requirements alongside. Nothing else starts until those two models exchange successfully.

Give the integration coordinator standing to make each partner write its side down.

What crosses each boundary, in what units, on which clock, and who owns it. Methods and code stay with the partner.

Accept a narrow demonstration.

One settlement configuration, one stress event, fully traceable. Not a platform, and not four couplings. This is the decision that costs you something.

In return, at week six.

That configuration under a normal case and a material stress event, every number traceable to the model, contract, schedule version and threshold set that produced it. And a test in week five that could tell you this recommendation was wrong.